Exploratory Data Analysis and Linear Regression for Student Academic Performance Analysis

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Abstract**—***Student academic performance prediction is a critical challenge in modern education. This study presents exploratory data analysis (EDA) and a linear regression-based prediction model applied to the UCI Student Performance Dataset (395 students). Attendance, study time, and period-based scores (G1, G2, G3) are analyzed to identify key performance factors. EDA findings confirm that prior assessment scores dominate final performance prediction, with G1–G3 and G2–G3 correlations of 0.80 and 0.90 respectively. The linear regression model achieves $R^2 = 0.847$ and MAE = 1.045 marks out of 20, demonstrating strong predictive accuracy. The system can serve as an early academic intervention tool enabling institutions to identify at-risk students after mid-term assessments.*

Keywords**—***Student Performance, Exploratory Data Analysis, Linear Regression, Education Analytics, Academic Prediction, Attendance Analysis.*

I. Introduction

The increasing availability of student data in educational institutions has created new opportunities for data-driven decision-making. Schools and universities routinely collect information such as attendance records, internal assessment scores, and study habits. However, a significant proportion of this data remains unanalyzed, limiting its potential to improve student outcomes.

Traditional approaches to evaluating student performance rely on end-of-semester grade reviews. This is reactive — by the time poor performance is identified, it is often too late for meaningful intervention. Predictive analytics and EDA offer a proactive alternative by flagging at-risk students before their performance deteriorates.

This paper applies EDA and linear regression to the UCI Student Performance Dataset [1], comprising 395 student records from a Mathematics course. Key research questions are: (1) Which features most significantly influence final marks? (2) Can a linear regression model reliably predict final marks from mid-term data?

II. Literature Review

A substantial body of research exists on student performance prediction. Table I summarizes key studies reviewed for this work.

TABLE I. Literature Review Summary

Author	Method	Result	Limitation
Cortez & Silva [1]	Regression & classification	G1/G2 are top predictors	Single subject
Romero & Ventura [2]	EDM survey	Clustering most useful	No unified framework
Baker & Inventado [3]	Learning analytics review	Early indicators help outcomes	LMS dependency
Bharadwaj & Pal [4]	Decision Tree (ID3)	Attendance & marks key factors	Small dataset
Yadav et al. [5]	NB, DT, Neural Network	Ensemble methods most accurate	High complexity
Marquez-Vera et al. [6]	Random Forest	88% dropout accuracy	One school only
Kabakchieva [7]	SVM & Logistic Reg.	SVM slightly outperforms LR	No feature ranking
Delen et al. [8]	DT, ANN, Logistic Reg.	DT highest accuracy	Imbalanced classes

A common finding across reviewed studies is that prior academic performance consistently outperforms attendance and study habits as a predictor. Most existing approaches use complex models (neural networks, ensembles) offering high accuracy but low interpretability. This study addresses that gap with a simple, interpretable linear regression model that achieves competitive accuracy and is practical for institutional deployment.

III. Problem Statement

Educational institutions collect large volumes of student data yet rarely analyze it systematically to support early academic intervention. Students at risk of poor final performance are typically identified only after grades have already deteriorated, leaving little time for corrective action.

The problem addressed is: which measurable student attributes most significantly influence final examination performance, and can a simple model reliably predict final marks using data available at the midpoint of an academic term? Current systems rely on manual review or basic statistics, which are insufficient for identifying complex patterns. An automated, data-driven, and interpretable approach is needed.

IV. Objectives of the Study

The specific objectives of this study are:

- To perform EDA on the UCI Student Performance Dataset to uncover patterns in student academic data.
- To examine the relationship between attendance, study time, and assessment-wise marks (G1, G2, G3).
- To identify features that most strongly correlate with final examination performance (G3).
- To build and evaluate a linear regression model for predicting final marks.

- To derive actionable insights that can inform academic intervention strategies.

V. Proposed Methodology

5.1 System Architecture

The proposed system follows a five-stage pipeline: (1) Data Collection — UCI dataset is loaded; (2) Data Preprocessing — features are cleaned and transformed; (3) EDA — visual and statistical analysis uncovers patterns; (4) Model Development — linear regression is trained; (5) Evaluation — model is assessed using R^2 and MAE metrics. The pipeline is lightweight, reproducible, and built entirely using open-source tools.

5.2 Algorithm Description

Linear regression models the relationship between a dependent variable and one or more independent variables. In this study, G3 (final marks) is predicted using G1, G2, study time, and attendance:

$$G3 = \beta_0 + \beta_1(G1) + \beta_2(G2) + \beta_3(studytime) + \beta_4(attendance) + \varepsilon$$

Coefficients are estimated using Ordinary Least Squares (OLS). The dataset is split 80% training and 20% testing. Figure 4 shows the model implementation code.

```
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_absolute_error, r2_score

# Select features and target
X = data[['studytime', 'attendance', 'G1', 'G2']]
y = data['G3']

# Split data into training and testing
X_train, X_test, y_train, y_test = train_test_split(X, y, te

# Create model
model = LinearRegression()

# Train model
model.fit(X_train, y_train)

# Predict
y_pred = model.predict(X_test)

# Evaluate model
print("Model Accuracy (R2 score):", r2_score(y_test, y_pred)
print("Mean Absolute Error:", mean_absolute_error(y_test, y_

Model Accuracy (R2 score): 0.8470256862898013
Mean Absolute Error: 1.0452298638783348
```

Figure 4. Linear Regression Model Implementation

5.3 Dataset Used

The UCI Student Performance Dataset [1] contains 395 records from a secondary-level Mathematics course in Portugal. Five attributes were selected: G1 (first internal exam, out of 20), G2 (second internal, out of 20), G3 (final exam — target), studytime (ordinal 1–4), and absences (converted to attendance %). The dataset was complete with no missing values. Figure 1 shows the data loading and preprocessing code.

```
[ ]
import zipfile

with zipfile.ZipFile("student.zip", 'r') as zip_ref:
  zip_ref.extractall("student_data")
print("Inner student.zip extracted successfully")

Inner student.zip extracted successfully

[ ]
import os
os.listdir("student_data")

['student-merge.R', 'student-mat.csv', 'student.txt', 'student-por.csv']

[ ]
import pandas as pd

data = pd.read_csv("student_data/student-mat.csv", sep=";")
# Create 'attendance' column from 'absences' before attempting to use it
data['attendance'] = 100 - data['absences']
# Now, select the desired columns and display their head
print(data[['attendance', 'studytime', 'G1', 'G2', 'G3']].head())

   attendance  studytime  G1  G2  G3
0          94          2   5   6   6
1          96          2   5   5   6
2          90          2   7   8  10
3          98          3  15  14  15
4          96          2   6  10  10
```

Figure 1. Data Loading and Preprocessing Code

5.4 Implementation Tools

The following tools were used in this study:

- Python 3.x — primary programming language
- Google Colab — cloud-based development environment
- Pandas — data manipulation and transformation
- Matplotlib and Seaborn — visualization and EDA
- Scikit-learn — model training and evaluation

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Create attendance from absences
data['attendance'] = 100 - data['absences']

# Create attendance groups
bins = [0, 80, 90, 101]
labels = ['<80%', '80-90%', '>90%']
data['attendance_group'] = pd.cut(data['attendance'], bins=bins, labels=labels, right

# Select required columns only
eda_data = data[['attendance', 'studytime', 'G1', 'G2', 'G3']]

# Show basic statistics
print(eda_data.describe())

# ----------------------------
# Attendance distribution
# ----------------------------
attendance_perf = data.groupby('attendance_group', observed=True)['G3'].mean()

plt.figure()
plt.bar(attendance_perf.index, attendance_perf.values)
plt.xlabel("Attendance Group")
plt.ylabel("Average Final Marks (G3)")
plt.title("Average Final Marks by Attendance Group")
plt.ylim(0, 20)
plt.xticks(rotation=0)

plt.show()
```

Figure 2. EDA and Visualization Code

```
# ----------------------------
# Study time distribution
# ----------------------------
study_perf = data.groupby('studytime')['G3'].mean()

plt.figure()
plt.bar(study_perf.index, study_perf.values)
plt.xlabel("Study Time Level (1:<2 hrs, 2:2–5 hrs, 3:5–10 hrs, 4:
plt.ylabel("Average Final Marks (G3)")
plt.title("Average Final Marks by Study Time Level")
plt.ylim(0, 20)
plt.show()


# ----------------------------
# Average marks
# ----------------------------
avg_scores = eda_data[['G1', 'G2', 'G3']].mean()

plt.figure()
plt.bar(avg_scores.index, avg_scores.values)
plt.xlabel("Assessment (G1: First Internal, G2: Second Internal,
plt.ylabel("Average Marks (Out of 20)")
plt.title("Average Student Marks Across Assessments")
plt.ylim(0, 20)
plt.show()

# ----------------------------
# Correlation heatmap
# ----------------------------
plt.figure()
sns.heatmap(eda_data.corr(), annot=True)
plt.title("Correlation Between Attendance, Study Time, and Marks"
plt.show()
```

Figure 3. Study Time, Average Marks, and Heatmap Code

VI. Results and Discussion

A. Attendance Distribution

Figure 5 shows the average final marks (G3) grouped by attendance category. Students in the >90% attendance group achieved the highest average G3 score (10.52), while the <80% group scored 10.29 and the 80–90% group scored 10.19. Results are summarized in Table II.

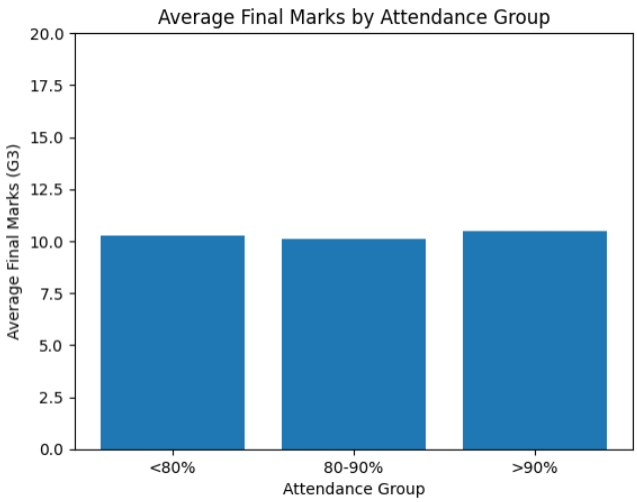

Figure 5. Average Final Marks by Attendance Group

TABLE II. Attendance Group vs. Average G3 Score

Attendance Group	Range	Avg. G3 Score
Low	< 80%	10.29
Medium	80–90%	10.19
High	> 90%	10.52

The narrow margin between groups indicates that attendance beyond a basic threshold yields diminishing returns on final performance. Attendance alone is not a reliable predictor of academic success.

B. Study Time vs. Final Marks

Figure 6 shows average G3 marks across study time categories. Students in category 3 (5–10 hrs/week) and 4 (>10 hrs/week) scored slightly higher, but the difference is modest. Academically weaker students may study more to compensate, which masks a stronger positive effect of study time.

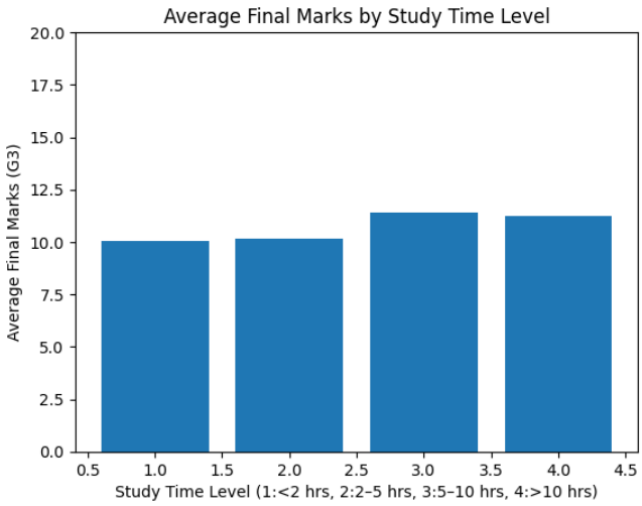

Figure 6. Average Final Marks by Study Time Level

C. Assessment-wise Average Marks

Figure 7 compares average marks across the three assessment periods: G1 ≈ 10.9, G2 ≈ 10.7, G3 ≈ 10.4. The slight downward trend may reflect increasing assessment difficulty or reduced engagement toward the end of term.

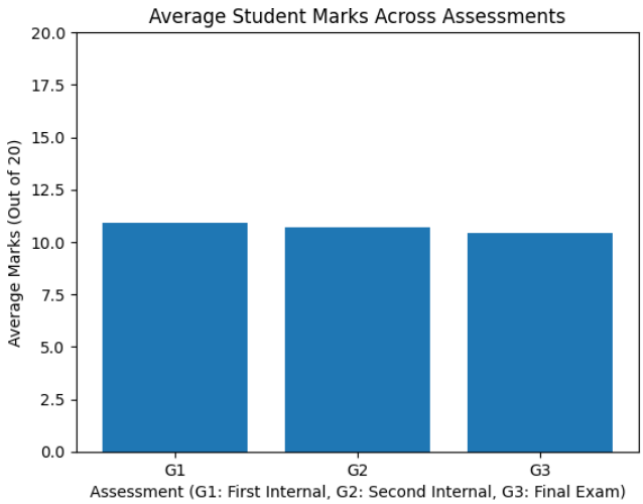

Figure 7. Average Student Marks Across Assessments (G1, G2, G3)

D. Correlation Analysis

Figure 8 shows the correlation heatmap. G2–G3 = 0.90 and G1–G3 = 0.80 are the strongest relationships. Attendance–G3 = −0.034 and Studytime–G3 = 0.098 are negligible. Prior assessment scores are the dominant predictors of final performance.

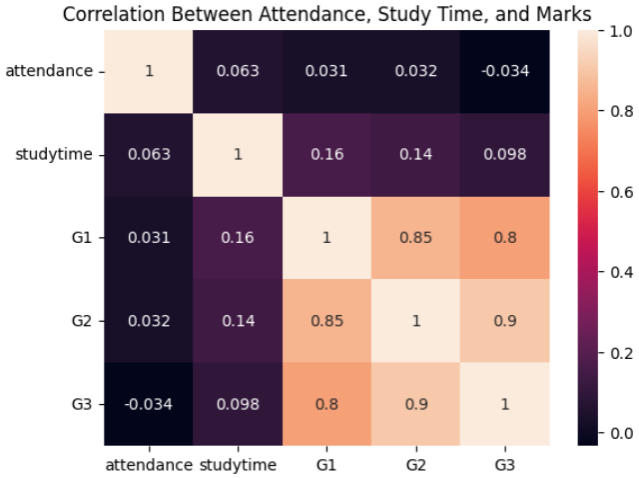

Figure 8. Correlation Heatmap: Attendance, Study Time, G1, G2, G3

E. Linear Regression Model Performance

Table III summarizes the model performance on the 20% test set. With $R^2 = 0.847$, the model explains 84.7% of variation in final marks. MAE = 1.045 confirms predictions are within approximately one mark of actual results — strong performance on a 20-point scale.

TABLE III. Linear Regression Model Performance

Metric	Value	Interpretation
R^2 Score	0.847	84.7% variance explained
MAE	1.045	~1 mark avg. error (out of 20)
G2–G3 Correlation	0.90	Very strong positive
G1–G3 Correlation	0.80	Strong positive
Att.–G3 Corr.	-0.034	Negligible
Study–G3 Corr.	0.098	Weak positive

VII. Applications

- Early Intervention: Deploy after G1 results to flag at-risk students and assign academic support before the final exam.
- Faculty Dashboards: EDA visualizations integrated into dashboards provide real-time class-level performance insights.
- Scholarship & Support Allocation: Predicted scores help administrators prioritize remedial support for borderline students.
- Policy Planning: Aggregated trends inform curriculum review and examination scheduling decisions.
- Student Self-Assessment: A student-facing tool can help learners estimate final performance based on current habits.

VIII. Advantages of the Proposed System

- Uses data institutions already collect — no new infrastructure required.
- Linear regression provides fully interpretable coefficients explaining each feature's contribution.
- EDA visualizations are accessible to non-technical educators without ML expertise.
- Applicable as early as after the first internal assessment (G1) for timely intervention.
- Computationally lightweight — runs on standard institutional servers without specialized hardware.
- Modular pipeline can be extended with additional features or advanced algorithms as data matures.

IX. Limitations

- Dataset is limited to one subject (Mathematics) from one Portuguese institution, reducing generalizability.
- Absences used as attendance proxy without verifying if they coincided with examination days.
- Linear regression may not capture complex non-linear student behavior patterns.
- Dataset excludes variables such as teacher quality, parental involvement, and socioeconomic status.
- Study time is ordinal (not continuous), limiting precision of its regression contribution.
- With only 395 records, the model may not generalize to larger or more diverse student populations.

X. Conclusion and Future Work

Conclusion

This study demonstrates the practical value of EDA and linear regression in predicting student academic performance. Analysis of the UCI Student Performance Dataset confirmed that prior assessment scores (G1 and G2) are the strongest predictors of final examination performance, with correlation values of 0.80 and 0.90 respectively. Attendance and study time showed weak independent effects.

The linear regression model achieved $R^2 = 0.847$ and MAE = 1.045, confirming strong predictive accuracy within a 20-mark system. These results support deployment of a simple, interpretable prediction model as an early academic intervention tool in educational institutions.

Future Work

Future extensions include: applying advanced algorithms such as Random Forest or Gradient Boosting to capture non-linear patterns; expanding the dataset across multiple subjects and institutions; integrating with a live LMS for real-time alerts; and incorporating socioeconomic and teacher-feedback variables for a more holistic model.

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